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A modular MgH_2 demonstration tank to store reversibly hydrogen

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ABSTRACT

While solid state hydrogen storage in metal hydrides has gained significant attention, the performance of built tanks depends strongly on their designs. Namely, attention must be paid to safety, energy efficiency, and implementation kinetics. Hence, the optimization of heat transfers remains a key point, considering the high enthalpy of the $Mg \leftrightarrow MgH_2$ reactions. Herein, the micro-sized MgH_2 powder was activated with additives during high energy ball milled, then blended with expanded graphite to optimize the heat transfers. The resulting disks after pressurization were enforced as stacks of flat capsules forming the here studied demonstration tanks. In fact, two original tank designs including MgH_2-filled capsules and intercalary heat-controller units were proposed. Correspondingly, two types of heaters were developed and numerically studied by using ANSYS Fluent codes. Thanks to large heat exchange surfaces set in between the elements forming the complete stacks, one of the proposed heating configurations was found to be more efficient promoting accordingly fast enough $Mg \leftrightarrow MgH_2$ reactions. On the other hand, subsequent experiments were carried out to ensure fast responses to the correlated thermodynamic conditions that are temperature and pressure as well as time-efficient absorption/desorption processes.

1. Introduction

Since global climate change has become a central concern, the search for new and clean energy resources has stimulated worldwide research activities accelerating innovation while restructuring energy industries. In parallel with the early merged renewable energy technologies such as photovoltaics [1] and wind power [2], the hydrogen vector appears as a promising clean and versatile solution. It would enable an effective shift from carbonaceous fuels to clean applications [3]. As energy carrier, hydrogen is expected playing a unique role in different sectors of our daily life such as electricity and heat productions [4,5], green ammonia manufacturing [6] and metallurgy [7].

However, one of the main challenges facing the hydrogen industry is related to storage issues because of its very weak density. In fact, storing hydrogen in liquid (LH) and gaseous (GH) forms [8] presents major drawbacks including high costs and significant safety issues [9]. For instance, LH storage requires extremely low temperatures (<20 K), while GH storage requires very high pressures (30–70 MPa). These routes need very specific and energy-consuming infrastructures, with increasing risks. Conversely, storing hydrogen in solid forms offers several notable advantages [10,11]. It was found safer and features high volumetric hydrogen densities. For example, Metal Hydrides (MH), allow compact and stable hydrogen storage, thus easing uses for various industrial and energy applications [4–7,12]. It is worthy to underline that solid state hydrogen storage (SSHS) solutions were actively developed. However, the optimization of hydrogen absorption/desorption (A/D) performance of (MH) tanks is vital for the development of safe energy storage solutions [9]. These tanks should enable an efficient storage by leveraging the capacity of hydrides to absorb and desorb hydrogen under specific temperature and pressure (T, P) conditions. In

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this context, many hydride materials have been studied owing to their ability to react with hydrogen in controlled manners. Particularly, $LaNi_5$ based alloys absorb and desorb hydrogen at moderate (T, P) conditions. They were used for storage applications as (NiMH) batteries, but the hydrides face limited storage capacity (<1.5 w%) [12]. Besides, the exotic PdH_x presents a unique ability to form at ambient pressure and temperature [13]. Unfortunately, its high cost prevents large-scale use. Conversely, magnesium hydride (MgH_2) is one of the most studied materials for hydrogen storage due to its high volumetric hydrogen density, its abundance and affordability [14–17]. However, its A/D conditions usually require temperatures higher than 300 °C [17]. Consequently, the rates of A/D processes were predominantly influenced by heat transfer and mass reactions within the tank. Therefore, enhancing the efficiency of heat and mass transfer in hydrogen hydride tanks is a key parameter for the development of this promising storage technique [18]. For this purpose, various approaches can be considered such as the enhancement of effective thermal conductivity using metal foams [19] or expanded natural graphite (ENG) [20–23]. Also, it was proposed to upgrade the heat transfer surfaces with fins, or integrate cooling tubes to implement the external heat transfers [24,25]. In our developed MgH_2 tank, we have chosen to incorporate ENG not only to enhance the thermal conductivity of the MgH_2 bodies but also to prevent agglomeration of the particles, thereby easing the hydrogen diffusion to reaction fronts. Besides, a better mechanical stability of the composite enables to increase its durability during the hydrogen A/D cycles. In this paper, novel heat exchange configurations are explored with the aim of optimizing the sorption process in a solid storage tank based on MgH_2. This current experimental research work is supported by numerical simulations to validate challenging high-performance tanks.

2. Metal hydride tank configurations

2.1. Proposed configurations

Metal hydride tanks offer a promising alternative for hydrogen storage in terms of efficiency and safety. However, their behavior and operation remain complex. The two essential elements for the tank's operation are the source and configurations of heating via heater elements, as well as the used hydrogen storage material.

2.1.1. A MgH_2 tank with a new design

The new designed tank was manufactured at JOMI LEMAN SAS. It consists of a stack of capsules as shown in Fig. 1. Each capsule contains two disks (diameter $\Phi = 30$ cm) of activated MgH_2. It is worth recalling that the primary MgH_2 was prepared from ~40 μm mean size of 98 w% pure Mg granules by using a dedicated temperature/pressure-controlled furnace (50 kg batch). The 7.3 H-w% powder was High-Energy Ball Milled (Zoz, 20 kg batch) for a few hours with TiVCr® [22] additive as catalyst.

Then the very reactive MgH_2 powder blended with ENG [21,22] in different proportions (here 4 w%). After pressing (1 ton/cm^2 by using a SATIL Corp. press), the resulting disks were A/D for up to 7400 cycles without loss in hydrogenation capacity. To assert the overall performances of the materials, Sievert's (PCT SETARAM Pro) and Differential Scanning Calorimetry (SETARAM 131) analyzers were used. Fig. 1 shows different parts (Fig. 1a–b) stacked as elemental bodies (Fig. 1c) of the tank, without using phase change materials (PCM) as was made elsewhere in order to optimize the energy management at the absorption/desorption (A/D) reactions via heat transfers with the PCM [26–28]. Herein, to simplify and to rend faster the demonstration unit, the exothermal reaction was controlled by an air-cooling system as explained hereafter. Finally, the MgH_2 based material was checked for its sorption kinetic characteristics at 320 °C with 6.5 w% absorbed in 10 min under 0.5 MPa (max. ~ 6.8 w%) and 6.0 w% desorbed in 17 min under 0.15 MPa (max. ~ 6.5 w%).

On Fig. 2, the assembly of the elements forming the new design unveils that each capsule shown in Fig. 1b was positioned between a steel heat diffuser (HD) corresponding to the previously described configuration and a disk equipped with an air cooler serpentine (2.6 m long). The heating elements were designed to heat the hydride material and reach the sorption temperature. Thermocouples, labelled R_1, R_2, R_3, and C_1, C_2, C_3, were positioned at different parts of the tank to control and measure temperatures. Similarly, other thermocouples, named S_1 and S_2, were placed aside the coolers. The latter were designed to reduce heat and prevent the temperature from exceeding a specific threshold, thereby preserving the properties of the hydride material used. The

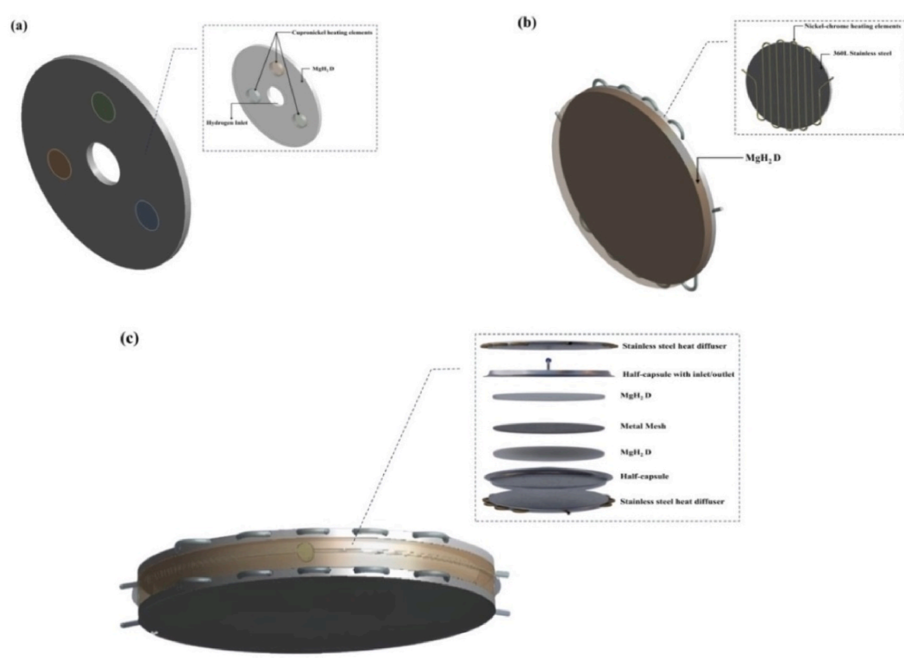

Fig. 1. The configurations: (a) (MgH_2-D) disk, (b) heat diffuser with its serpentine heat resistance (2.6 m long), (c) a steel (316 SS) capsule containing two (MgH_2-D) disks in between two heater-diffusers.

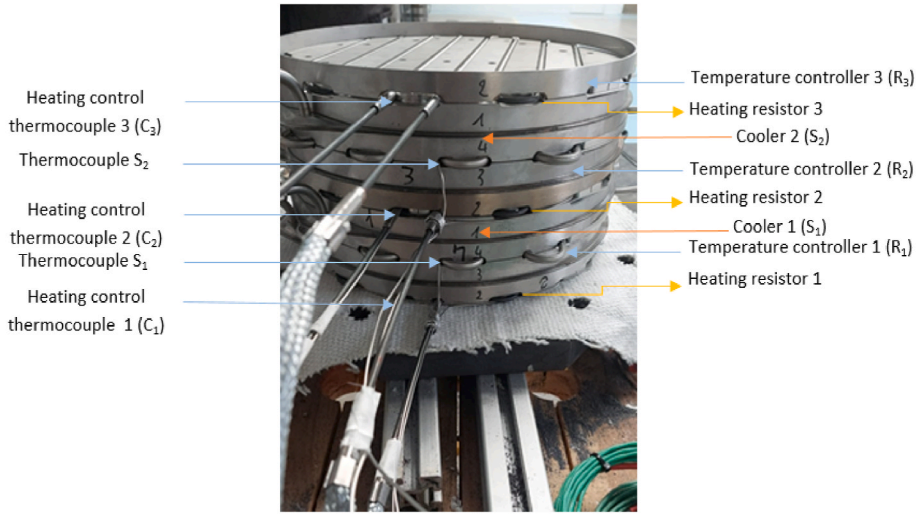

Fig. 2. The newly developed magnesium hydride tank.

present tank also features thermal insulation surrounding the entire vessel, thus minimizing thermal losses. Furthermore, flanges were fixed at the top and bottom of the tank to secure the various components.

2.1.2. Numerical simulations and experimental strategies

In this study, the heating element corresponds to a stainless-steel heat diffuser. Additionally, magnesium hydride-containing disks were designed for hydrogen storage. Throughout this study, both simulations and experiments are used to investigate the effect of different proposed heat diffuser configurations on the thermal behavior and A/D performance of above designed MgH_2 tank.

In this way, two configurations were numerically studied. The first one, as shown in Fig. 1a, features an activated (MgH_2-D) with perforations designed to house three disk heaters. These heating elements were included to heat the disk at the set temperature for the hydrogen A/D procedures. This setup represents the standard approach. In the second configuration, the heating elements were integrated into the steel heat diffuser (HD) arranged in a serpentine pattern. The heat diffuser was then connected to the (MgH_2-D) unit, as shown in Fig. 1b. This enables us to maximize the contact surface between the heating element and the disk, ensuring a better heat dispatch. Finally, the study extends to consider the impact of stainless-steel (SS) capsules that contains two (MgH_2-D) separated by a metal mesh (MM), as depicted in Fig. 1c. This capsule was fixed at both the upper and lower faces to the heat diffuser shown in Fig. 1b. Consequently, the test plan begins with a desorption step, according to both the numerical simulation and experimental investigations for all the further processes.

The whole series of experimental tests carried out on the manufactured metal hydride tank have consisted of the following steps. First, a set-point temperature was established depending on the selected absorption or desorption phase. For the absorption test, the set-point was 340 °C, while 360 °C is settled for desorption according to previous simulations [23,27,29] and tests conducted on other tank designs [25,30,31].

The desorption test started with a preheating to 150 °C with a power rating of 750 w in each heater, for a total of 1500 W (each disk being in contact with two heaters). Typically, the preheating temperature should be close to the set-point temperature. In our case, we opted for a lower temperature to assess the thermal diffusivity, thus the time required to reach the target temperature. Then, the preheating for an additional 2 h while maintaining a temperature of 360 °C heating was paused, followed by a desorption process, completed within ~3 h. The absorption process was initiated 12 h after the desorption test noticing that thanks to the thermal insulation, the tank is cooled down and then maintained at 210 °C. This has allowed the initiation of the absorption without preheating. Subsequently, the temperature began to rise due to the enthalpy of formation (exothermic reaction) in accordance with Eq. (1). To maintain the reaction rate, the excess heat was evacuated accordingly by using the air-cooling circuit, which is regulated via the distributed thermocouples throughout the structure.

$$Mg + \frac{X}{2}H_2 \leftrightarrow MgH_2 + \Delta H \,(\Delta H \sim 74\, kJ\,/\,mole\ \mathrm{H_2}) \tag{1}$$

2.2. Mathematical thermal transfer calculation

2.2.1. Mathematical model description

The simulation studies were conducted using ANSYS Fluent [32]. For the first configuration the computation considers an activated (MgH_2-D) with three disk heaters as shown in Fig. 3a. For the second configuration the computation includes a steel heat diffuser (HD) as illustrated in Fig. 1b in contact with an activated (MgH_2-D). Both cases were found governed by the Fourier equation of heat propagation, as concerning the heating elements, the steel (HD), and the (MgH_2-D). Finally for this configuration the computation includes the entire system as depicted in Fig. 1c.

For all these different configurations, the (MgH_2-D) containing capsule was filled with hydrogen, meaning that a convection term was to be considered. A representative model detailing the different concerns of heat transfer is shown in Fig. 3.

The geometrical and thermal properties of each component in the studied configurations are presented in Table 1, while the thermophysical properties of heating element, (MgH_2-D) and hydrogen, were obtained from Refs. [21,24,27,28]. All the parameters of columns 1 and 3 are defined in the Table Nomenclature.

2.2.2. Thermal transfer calculations

Analysis of the first configuration presented in Fig. 3a includes four nodes: the heater element, hydrogen gas, (MgH_2-D) and insulation cover, labelled as disk heater, H_2, (MgH_2-D), and I, respectively. Following, the second model shown in Fig. 3b includes five nodes: the stainless-steel tube, heater, heat diffuser, (MgH_2-D), and hydrogen gas, respectively labelled as SS, Cu–Ni, (MgH_2-D), and H_2. Based on the energy balance equation (2), a set of nonlinear equations were derived governing the temperature distribution in each of the components. The different thermal variables are T_{HCE}, T_{H_2} $T_{MgH_2\ D}$, T_i, T_{ss}, T_{ni-ch}, and T_{HD} considering the conduction and convection heat transfer mechanisms.

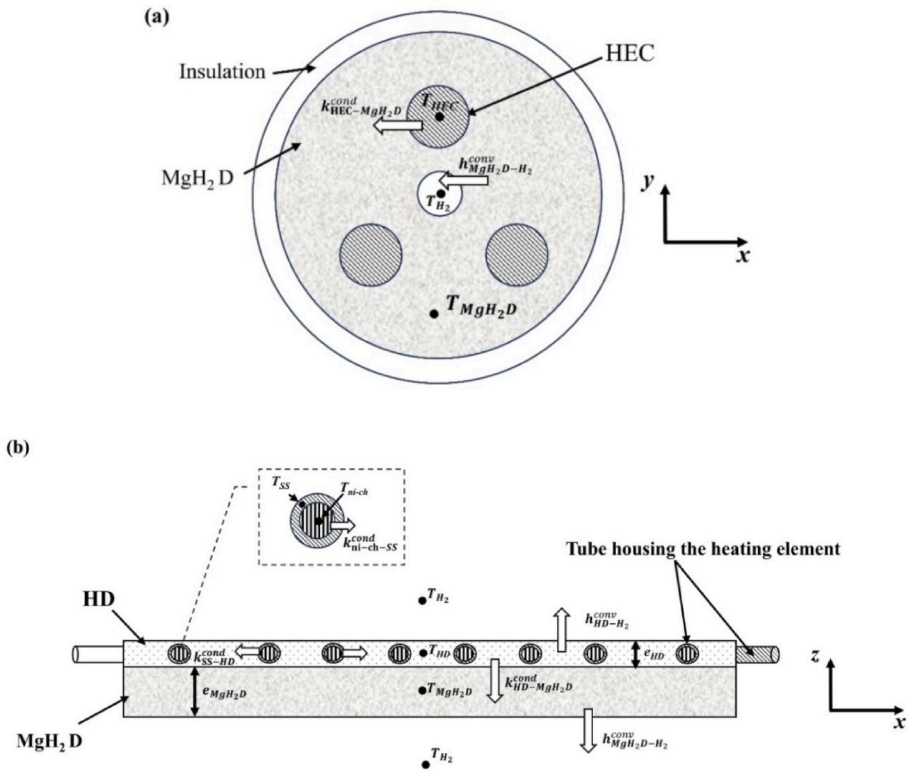

Fig. 3. Designs of the first studied configuration 1: (a) (MgH_2-D) with three disk heaters, (b) (HD) with a serpentine temperature control element fixed to the (MgH_2-D).

Table 1
Properties of each component considered for the numerical study.

Heating element		(MgH_2-D) unit	
Parameters	Values	Parameters	Values
ρ_{HCE}	8940 kg/m^3	$\rho_{\mathrm{MgH_2\ D}}$	1800 kg/m^3
Cp_{HCE}	377 J/(kg·K)	$Cp_{\mathrm{MgH_2-D}}$	1545 J/(kg·K)
e_{HCE}	0.01 (m)	$A_{\mathrm{MgH_2\ D}}$ (config.1)	0.28 (m^2)
r_{HCE}	0.021 (m)	$e_{\mathrm{MgH_2\ D}}$ (config.1)	0.015 (m)
k_{HCE}	40 W/(m·K)	$k_{\mathrm{MgH_2\ D}}$	2 W/(m·K)
ρ_{ni-ch}	7920 (kg/m^3)	$A_{\mathrm{MgH_2\ D}}$ (config.2)	0.28 (m^2)
Cp_{ni-ch}	460 J/(kg·K)	$e_{\mathrm{MgH_2\ D}}$ (config.2)	0.015 (m)
l_{ni-ch}	0.15 (m)	$\varepsilon_{\mathrm{MgH_2\ D}}$	0.3
r_{ni-ch}	0.0025 (m)	Hydrogen	
k_{ni-ch}	13.18 W/(m·K)	Parameters	Values
ρ_{HD}	8000 kg/m^3	$r_{\mathrm{H_2}}$	0,06 (m)
Cp_{HD}	500 J/(kg·K)	$T_{\mathrm{H_2}}$	300 (K)
e_{HD}	0.0085 (m)	$K_{\mathrm{H_2}}$	0.24 W/(m·K)
A_{HD}	0.28 (m^2)	$Cp_{\mathrm{H_2}}$	14000 J/(kg·K)
k_{HD}	14 W/(m·K)		
ρ_{SS}	8000 kg/m^3		
Cp_{SS}	500 J/(kg·K)		
l_{SS}	0.0085 (m)		
r_{SS}	0.15 (m)		
k_{SS}	14 W/(m·K)		

$$\rho_j Cp_j V_j \frac{dT_j}{dt} = Q_{j,in} - Q_{j,out} + Q_{j,gen} \tag{2}$$

where $\rho_j Cp_j V_j$ represent the thermal power stored in the studied volume. ρ_i, Cp_i, V_i refer respectively to the density, specific heat and volume of the component j. Q_{in}, Q_{out} and Q_{gen} represent respectively the incoming, outing and generated thermal energies within the studied volume.

2.2.2.1. Configuration 1. The (Cu–Ni) heater combines excellent electrical resistance with high thermal conductivity. The heater conversion of electricity (*HCE*) to heat ($Q_{HCE,gen}$) was expressed by I^2R where I and R are the electrical current and the resistance, respectively. The direct contact between the heater and the (MgH_2-D) eases the heat transfer ($q^{cond}_{HCE-\mathrm{MgH_2\ D}}$) and represents the main source of energy for the system, Thus, the heat balance equation for the HCE can expressed as follows:

$$\rho_{HCE} Cp_{HCE} V_{HCE} \frac{dT_{HCE}}{dt} = Q_{HCE,gen} - q^{cond}_{HCE-\mathrm{MgH_2\ D}} \tag{3}$$

For the (MgH_2-D), heat exchange occurs by conduction from the heater (HCE) ($q^{cond}_{HCE-\mathrm{MgH_2\ D}}$), while convective heat exchange takes place between the (MgH_2-D) body and hydrogen ($q^{conv}_{\mathrm{MgH_2\ D-H_2}}$). In this case, the heat balance equation for the (MgH_2-D) can be written as follows:

$$\rho_{\mathrm{MgH_2}\ D} Cp_{\mathrm{MgH_2}\ D} V_{\mathrm{MgH_2}\ D} \frac{dT_{\mathrm{MgH_2}\ D}}{dt} = q^{cond}_{HCE-\mathrm{MgH_2\ D}} - q^{conv}_{\mathrm{MgH_2\ D-H_2}} \tag{4}$$

Regarding the heat balance of hydrogen gas, the rejected heat from the (MgH_2-D) was transferred by convection ($q^{conv}_{\mathrm{MgH_2\ D-H_2}}$).

$$\rho_{\mathrm{MgH_2}\ D} Cp_{\mathrm{MgH_2}\ D} V_{\mathrm{MgH_2}\ D} \frac{dT_{\mathrm{MgH_2}\ D}}{dt} = q^{cond}_{HCE-\mathrm{MgH_2\ D}} - q^{conv}_{\mathrm{MgH_2\ D-H_2}} \tag{5}$$

Concerning the insulation layer, no heat transfer was considered ($\rho_i Cp_i V_i \frac{dT_i}{dt} = 0$).

2.2.2.2. Configuration 2. The (Ni–Cr) new *HCE* has the same function as here above being the primary heat source for the system ($Q_{HCE,gen}$). The heater in direct contact with the stainless-steel tube (SS) ($q^{cond}_{HCE-\mathrm{MgH_2\ D}}$) impacts the corresponding heat balance as follows:

$$\rho_{ni-ch} Cp_{ni-ch} V_{nHCE} \frac{dT_{ni-ch}}{dt} = Q_{ni-ch,gen} - q^{cond}_{HCE-\mathrm{SS}} \tag{6}$$

In a second layer, conduction heat occurs between the steel tube (SS) and the heat diffuser (HD) ($q^{cond}_{SS-(\mathrm{HD})}$). The balance equation for the steel tube (SS) is accordingly given by:

$$\rho_{SS} Cp_{SS} V_{SS} \frac{dT_{SS}}{dt} = q^{cond}_{HCE-SS} - q^{cond}_{SS-(HD)} \quad (7)$$

The diffuser (HD) recovered heat from the steel tube (SS) then transferred both by conduction to the (MgH_2-D) unit. Thus, the heat balance equations for the (HD) and the (MgH_2-D) can be written as:

$$\rho_{HD} Cp_{HD} V_{HD} \frac{dT_{HD}}{dt} = q^{cond}_{SS-(HD)} - q^{cond}_{)-MgH_2\ D} \quad (8)$$

$$\rho_{MgH_2\ D} Cp_{MgH_2\ D} V_{MgH_2\ D} \frac{dT_{MgH_2\ D}}{dt} = q^{cond}_{(HD)-MgH_2\ D} - q^{conv}_{MgH_2\ D-H_2} \quad (9)$$

Detailed expressions for each thermal flux were listed in Table 2, where the heat transfer coefficients $\left(h^{conv}_{MgH_2\ D-H_2}\right)$ were calculated as follows:

$$h^{conv}_{MgH_2\ D-H_2} = \frac{k_{H_2} * Nu_{MgH_2\ D-H_2}}{r_{H_2}} \quad (10)$$

where the Nusselt number $Nu_{MgH_2-D-H_2}$ was determined from:

$$Nu_{MgH_2\ D-H_2} = C \cdot Re^n_{H_2} \quad (11)$$

Since the convection regime is laminar, the value of C and the coefficient n are calculated as follows:

$C = 0.59$, $n = 1/4$ for $10^4 < Re < 10^7$ (laminar regime) with: $Re = \frac{g\beta\ (T_{MgH_2\ D} - T_{H_2}) D^3_{H_2}}{\nu\ \alpha}$

where g, β, ν, α represent gravity, coefficient of thermal expansion of hydrogen, diameter of hydrogen passage cavity, viscosity of hydrogen and thermal diffusivity of hydrogen, respectively.

To analyze the thermal performance of the system, the governing equations of heat transfer (Eqs. (3)–(9)) were solved numerically using the finite volume method (FVM), as implemented in ANSYS Fluent. The computational domain was discretized into a structured mesh, ensuring adequate refinement in regions with high thermal gradients. The implemented algorithm was employed for pressure-velocity coupling, ensuring convergence of the numerical solution. The boundary conditions defined in Table 3 were applied to simulate realistic operating conditions, and a grid independence study was conducted to validate the accuracy of the numerical model. Convergence was found achieved when residuals for energy drop below 10^{-6}, ensuring a reliable solution for further analysis.

3. Results and discussion

3.1. Numerical simulation

The heating phase plays a crucial role in triggering the sorption process within the MgH_2 hydride material by initiating the sorption reactions of hydrogen. It directly affects the temperature of the materials, which in turn influences the fastness and efficiency of the process. Optimized thermal management improves the performance of the (MgH_2-D) by easing the absorption and the release of hydrogen under specific temperature and pressure conditions. Consequently, the heating time of the (MgH_2-D) influences the overall heating of the metal hydride tank. Rapid and efficient (MgH_2-D) heating will result in a quick response from the tank, enabling faster hydrogen absorption or desorption. Conversely, slow heating will delay the processes. This relationship between the (MgH_2-D) heating rate and the tank heating underscores the importance of well-designed thermal management to optimize the efficiency of the hydrogen storage system. To reach the desired temperature of the (MgH_2-D) units we propose a new configuration featuring a stainless-steel heat diffuser, in which the heating elements were integrated in the form of a coil. This configuration was implemented after simulation results demonstrated its effectiveness compared to the standard configuration, where the heating elements were arranged as tubes positioned inside of the (MgH_2-D) units.

Table 3
Imposed boundary conditions for the numerical simulation.

Initial Conditions	Boundary Conditions
• Initial temperature of MgH_2-D was 300 K • Initial temperature of heat diffuser was 300 K	• Outer wall of tank was thermally insulated
• Temperature of heating element was 633 K	• Hydrogen inlet was at temperature of 300 K

Simulation results of the effect of the applied heating configurations are presented separately. Figs. 4–6 illustrate the thermal behavior of the (MgH_2-D) under different heating conditions. In the standard configuration illustrated in Fig. 4, the time required to heat the (MgH_2-D) was significantly long. As shown in Fig. 4c, the temperature of the (MgH_2-D) units rises from 300 to 400 K within 9 min, but this thermal increase remains confined to the regions near the heating element. However, achieving a uniform temperature distribution across the entire disk requires a much longer duration, exceeding 180 min, as depicted in Fig. 4d. This figure highlights a more extensive thermal propagation across the disk but without reaching complete uniformity over its entire surface, confirming that the time required for overall heating was high.

On one hand this phenomenon was explained by the ratio between the size of the heating element and the surface area to be heated. Specifically, the (MgH_2-D) surface was relatively large compared to the diameter of the heating element, which limits the heat transfer efficiency and slows down thermal diffusion throughout the disk. This structural constraint results in a pronounced thermal gradient, where only the areas in direct contact with the heating element rapidly reach the desired temperature, while the rest of the disk experiences a much slower temperature rise.

On the other hand, although Cu–Ni is a material with high thermal conductivity, the heating disks were positioned at only three specific locations on the (MgH_2-D). This limited configuration restricts contact with the entire surface of the disk, preventing uniform heat transfer. Furthermore, the heat generated by the heating elements was concentrated in these specific areas, creating significant thermal gradients between the heated and non-heated regions. Moreover, this configuration may lead to localized overheating, which could affect the stability of the MgH_2 material and its ability to optimally absorb and desorb hydrogen.

In the proposed new configuration, a steel Heat Diffuser (HD)

Table 2
Thermal fluxes for the studied configurations.

Configuration 1(Fig. 3a)	Configuration 2 (Fig. 3b)
$Q_{HCE,gen} = I^2R\ q^{cond}_{HCE-MgH_2\ D} = 2\pi e_{HCE} k^{cond}_{HCE-MgH_2\ D}(T_{HCE} - T_{MgH_2\ D})/\log\left(\frac{r_{MgH_2\ D}}{r_{HCE}}\right)$	$Q_{HCE,gen} = I^2R\ q^{cond}_{HCE-SS} = 2\pi l_{HCE} k^{cond}_{HCE-SS}(T_{HCE} - T_{SS})/\log\left(\frac{r_{SS}}{r_{HCE}}\right)$
$q^{conv}_{MgH_2\ D-H_2} = 2\pi r_{H_2} e_{MgH_2\ D} h^{conv}_{MgH_2\ D-H_2}(T_{MgH_2\ D} - T_{H_2})$	$q^{cond}_{SS-(HD)} = \frac{k^{cond}_{SS-HD} * A_{HD}}{e_{HD}}(T_{SS} - T_{(HD)})$
	$q^{cond}_{(HD)-MgH_2\ D} = \frac{k^{cond}_{HD-MgH_2\ D} * A_{MgH_2\ D}}{e_{MgH_2\ D}}(T_{(HD)} - T_{MgH_2\ D})$
	$q^{conv}_{MgH_2\ D-H_2} = h^{conv}_{MgH_2\ D-H_2} * A_{MgH_2\ D}(T_{MgH_2\ D} - T_{H_2})$

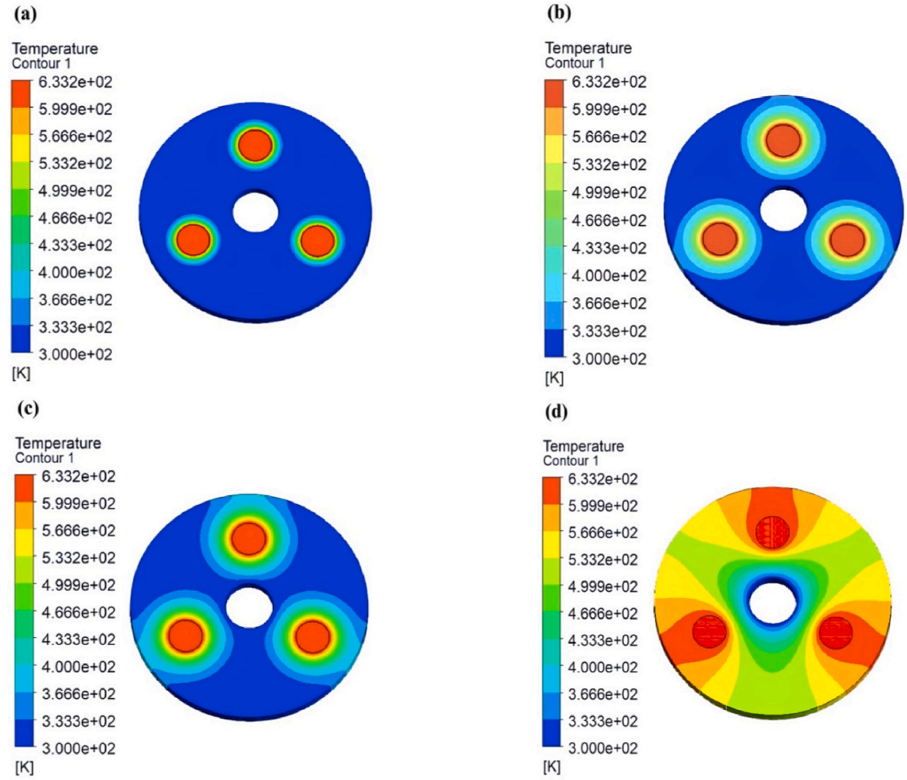

Fig. 4. Effect of the disk heaters (**configuration 1**) on the (MgH_2-D) unit over time: (a) **1 min**, (b) **6 min**, (c) **9 min**, (d) **180 min.**

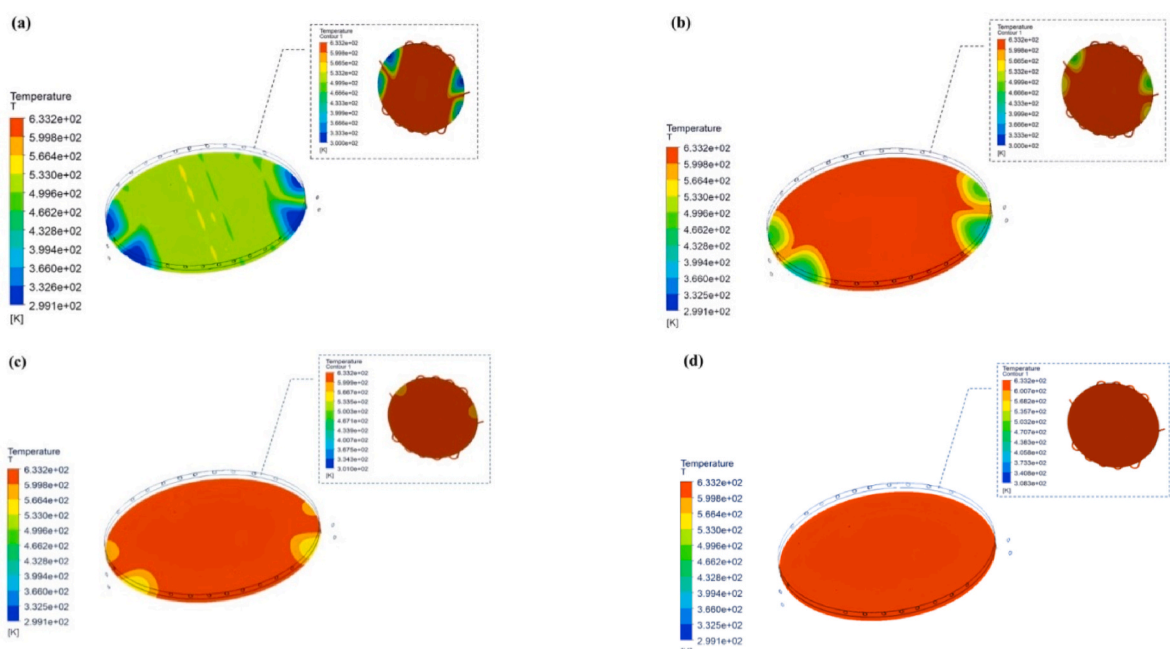

Fig. 5. Cross-section view illustrating the heating effect (configuration 2) on the (MgH_2-D) under the impact of the serpentine heater **configuration 2** over time (a) **1 min**, (b) **3 min**, (c) **6 min**, (d) **9 min.**

incorporates a serpentine heating element (Fig. 1b), designed to maximize thermal contact with the (MgH_2-D) unit. As a result, the (HD), heater was in direct contact with the (MgH_2-D) unit enabling uniform and rapid heating of the disk in just 9 min. The temperature distribution within the (MgH_2-D) is illustrated in Fig. 5, which presents a cross-section in the XY plane, highlighting the radial and axial propagation of heat throughout the disk. The thermal diffusion initially results in a high temperature gradient near the (HD), followed by gradual homogenization due to the thermal conductivity of (MgH_2-D) and internal heat exchanges. After a few minutes, the entire disk reaches a stable temperature, thereby optimizing the efficiency of the hydrogen storage and release process. The (Ni–Cr) heater was chosen for its high mechanical strength and durability in high-temperature environments and so a longer lifetime. The heating process relies on progressive thermal

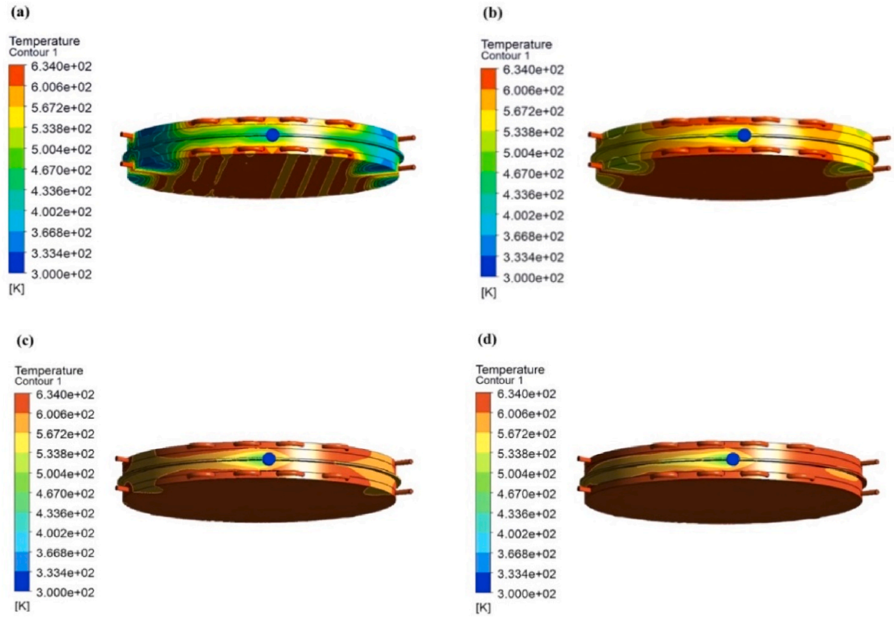

Fig. 6. Heating **configuration 2** of a capsule over time: (a) **1 min**, (b) **3 min**, (c) **6 min**, (d) **29 min.**

dissipation, where the serpentine heating element rapidly increases the temperature of the steel (HD) from 300 to 533 K within 1 min, as shown in Fig. 5a. This diffuser plays a crucial role in homogenizing the heat flux by efficiently transferring thermal energy to the (MgH_2-D) unit which in turn, reaches 257 °C in just 1 min.

Compared to the previous configuration 1, this design demonstrates significant improvements in (HD) efficiency and uniformity. The direct contact between the (HD), which incorporates a serpentine heating element, and the (MgH_2-D) eliminates thermal impedances existing in the previously discussed configuration, leading to a faster and more homogeneous temperature distribution. This enhances the overall performance of the hydrogen absorption and desorption process. Regards to the heat distribution within the (MgH_2-D) in straight contact with the (HD), we extended our analysis to assess its impact on the entire capsule designed to accommodate two (MgH_2-D). The capsule serves as a base unit then replicated in the design of the solid-state hydrogen storage demonstrator. Ensuring uniform and efficient heat propagation throughout the entire capsule was crucial to optimize the hydrogen sorption processes of the reservoir.

As illustrated in Fig. 6, which represents the evolution of heat propagation across the entire capsule, this one reaches the required temperature of 630 K after 29 min as shown in Fig. 6d. It was noticeably longer than the 9 min needed to heat a single (MgH_2-D). This difference can be explained by several factors related to the design of the demonstrator. The capsule contains two (MgH_2-D) units along with a superstructure as support which increases both the sensible heat to provide and the thermal inertia. Furthermore, while the capsule was insulated, its geometry differs from that of the single disk. The two (MgH_2-D)s were separated by a diffuser, so the heat transfer is made more complex.

Despite the increase in heating time, the here considered design remains well efficient in terms of heat distribution and process time. Additionally, it allows better control of the parameters influencing the hydrogen absorption and desorption steps such as H_2 pressure and dissipation of heat generated by the reaction. It was also worth noting that the capsule included a cooling tube within the diffuser channel, activated when the reaction temperature exceeds a predetermined threshold.

3.2. *Absorption/desorption experiments*

A fourfold unit capsules was supported forming an axial staking structure, as shown Fig. 2. The completed demonstrator was experimentally tested to evaluate the absorption and desorption processes. Following the preheating time (-0h03), for desorption the thermal monitors R_1, R_2 and R_3 were set at ~360 °C while accordingly the thermocouples C_1, C_2 and C_3 show 339–357 °C. Thirty minutes later, the mass-flow showed a gas outing rate of ~99 g/h, as the tank pressure decreases from 6 to 4.1 bar.

The end of the desorption process takes place roughly ~3:45 h later, as shown in Fig. 7. During this a mass of 362.5 g of hydrogen was desorbed as shown on Table 4. Furthermore, the four capsules in the tank reached a temperature ranging between 340 and 360 °C, which corresponds to the range indicated in the literature, thus confirming effective desorption. As is well known, for the absorption test the reaction process begins immediately since the tank was maintained at a high temperature enough after the end of the desorption test. As shown in Fig. 8 and

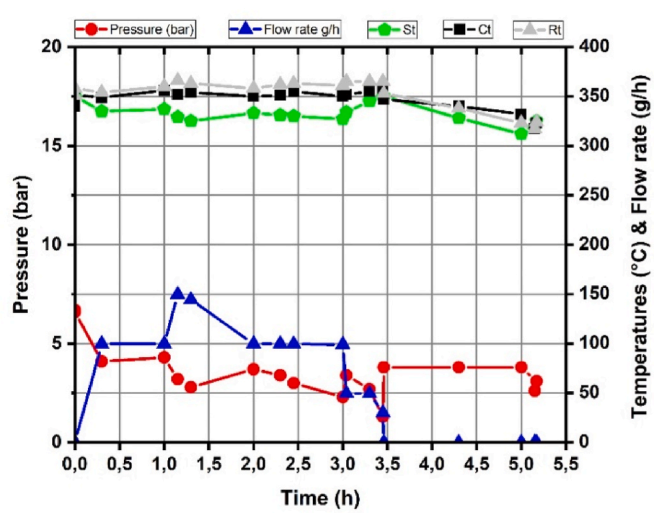

Fig. 7. Behavior of temperature, pressure and flow rate versus time during the hydrogen desorption. The lines are guides to the eyes.

Table 4
Results recorded step by step during the desorption process (S, C, R are given in Fig. 2).

Time	S_1 (°C)	S_2 (°C)	C_1 (°C)	C_2 (°C)	C_3 (°C)	R_1 (°C)	R_2 (°C)	R_3 (°C)	Flow rate g/h	Pressure bar
−0:03	351	350	337	353	331	349	356	349	0 (0)	6.7
0:00	351	349	349	355	341	357	359	359	0 (100)	6.6
0:30	335	336	353	357	339	355	355	352	99.7 (100)	4.1
1:00	337	337	362	358	349	367	356	359	99.7 (100)	4.3
1:15	329	329	356	353	352	370	367	361	149.6 (150)	3.2
1:30	324	325	352	357	354	368	357	366	144.3 (150)	2.8
2:00	333	333	352	352	348	356	359	359	99.7 (100)	3.7
2:30	331	332	354	351	348	367	363	356	99.7 (100)	3.4
2:45	329	330	353	355	358	358	365	367	99.6 (100)	3
3:00	326	329	350	350	352	359	359	365	99.0 (100)	2.3
3:04	333	335	351	352	354	362	365	368	49.6 (50)	3.4
3:30	339	351	–	351	360	–	–	–	49.5 (50)	2.7
3:45	345	353	–	353	363	–	–	–	30 (50)	1.3
3:46	347	354	349	346	348	351	351	357	0	3.8
4:30	325	331	340	337	344	331	339	344	0	3.8
5:00	309	316	333	326	338	321	315	335	0	3.8
5:17	324	326	322	324	325	322	325	326	0	3.1
5:15	317	319	315	318	317	315	319	318	0	2.6

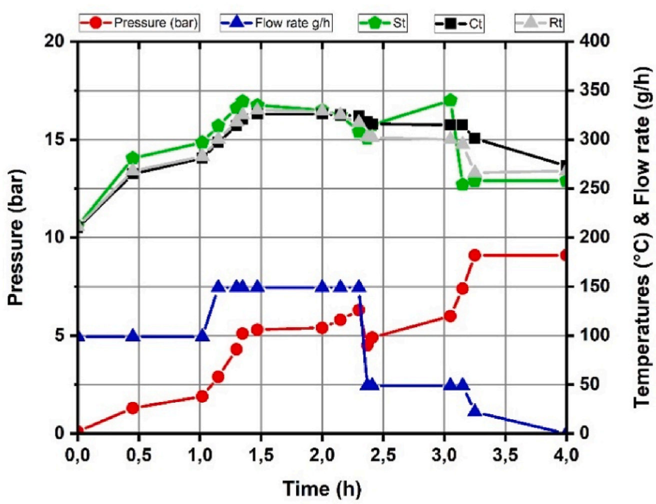

Fig. 8. Behavior of temperature, pressure and flow rate versus time during hydrogen absorption. The lines are guides to the eyes.

Table 5, upon initiating the test and opening the valves, an hydrogen absorption rate of ~100 g/h under a pressure of 0.1 bar was measured.

During the storage process, the temperature and the pressure of the tank continue to increase with the amount of hydrogen absorbed. Hene, the temperature reaches values above 340 °C only 1.5 h later, particularly in the center of the stack (thermocouple C_2). This prompted us to implement cooling regulations to control the temperature and prevent the end of the absorption reaction because of the phase equilibrium limit $Mg \leftrightarrow MgH_2$ [17,33]. It is worth underlining that during the test, the set-point flow rate of the incoming hydrogen was experimentally decreased from ~150 to ~50 g/h. This immediately resulted in tank cooling since the hydrogen reaction was reduced. Alos, this has affected more particularly the C_3 and R_3 thermometers of the external and basal capsule in contact with the metallic table, i.e. the less insulated. However, a counter response was automatically delivered by the electronic controller to impulse the internal pressure delivered to the tank. Such an experiment demonstrates the prompt reactivity of this newly designed tank.

At the end of the absorption test, the Mg to MgH_2 transformation was made complete, the inning pressure level was increased, the H_2 flow vanished, and all the measured temperatures begin to slightly decrease. The mass flowmeter recorded a total absorption of 364 g in very fair agreement with the desorption capacity of 362.5 g, i.e. a tank capacity >4 Nm^3.

3.3. *Efficiency estimates*

Even though the newly designed tank with heater configuration 2, has not been fully optimized, an estimate of its energy storage efficiency was carried out. For this purpose, it is considered first that the tank was maintained continuously working at the reaction temperature of

Table 5
Results recorded step by step during hydrogen absorption process (S, C, R are given in Fig. 2).

Time	S_1 (°C)	S_2 (°C)	C_1 (°C)	C_2 (°C)	C_3 (°C)	R_1 (°C)	R_2 (°C)	R_3 (°C)	Flow rate g/h	Pressure bar
0:00	210	213	208	211	211	208	213	212	99.6	0.1
0:45	285	277	263	286	246	268	287	249	99.7	1.3
1:02	300	294	282	300	263	282	302	267	99.7 (100)	1.9
1:15	317	311	298	316	277	304	318	279	149.8 (150)	2.9
1:30	334	330	316	333	294	323	335	297	149.7 (150)	4.3
1:35	341	338	322	340	301	330	342	305	149.7 (150)	5.1
1:47	337	334	327	345	307	335	345	311	149.7 (150)	5.3
2:00	333	328	327	344	308	333	345	310	149.7 (150)	5.4
2:15	333	318	326	346	304	329	346	301	149.7 (150)	5.8
2:30	298	267	326	350	297	318	347	288	149.7 (150)	6.3
2:37	318	285	321	343	291	303	330	278	49.6 (50)	4.5
2:41	330	299	318	343	289	302	328	277	49.6 (50)	4.9
3:05	300	281	313	348	286	301	323	277	49.6 (50)	6
3:15	262	247	311	352	284	296	314	275	49.6 (50)	7.4
3:25	267	250	294	338	273	268	270	260	22.4 (50)	9.1
4:00	275	267	278	277	265	274	270	262	0 (50)	9.1

approximately 320 °C, while, the absorption and desorption processes were operated by external pressures of 1.5 and 0.2 MPa, respectively. The stack forming the tank was insulated with a 10 cm thick Rockwool-035® blanket leading to an insulation index of ~0.8. So, during a sorption cycle, the energy loss of sensible heat ΔLS which is mainly due to the mass of steel superstructure, was approximated as ~3.2 kWh. On the other hand, the hydrogen energy stored in the tank corresponds to HE = ~12 kWh (~0.364 kg H_2). Accordingly, the heat of MgH_2 formation (ΔS) to be outed during the absorption process accounts for ~1/3 HE (74 kJ/mol H_2). So, the present energy storage efficiency may be approximated as ESE = (HE-ΔS-ΔLS)/(HE-ΔS) ~ 0.6. It might be up to 0.75 if reversible heat storage was leaned in the form of a PCM tank ancillary.

Once again, better performances can be expected with a less mass of steel and if better thermal insulation are considered. Conversely, the values of ESE should be severely reduced if the tank is stopped for a long time. In matter of SSHS tanks, it is challenging to design configurations enabling to transfer and recover the heat of H – M reaction. In ref. [25] a recent review on heat exchanger geometry was made, however mainly addressed to low temperature metal hydrides. The large review on magnesium-based materials for hydrogen-based energy storage [16] gathers plenty of interesting information on the various characteristics of the base MgH_2 materials. Also, the latter paper cites a panel of potential applications including H-reversible storage, heat transfer, heat production, energy and regeneration via SOF, while indicating the corresponding tanks and ancillaries. Nevertheless, to our knowledge the, only development at industry level of activated MgH_2 powders (up to 30 t/year) followed by various types and scales of equipped tanks was detailed in ref. [34]. However, the new demonstrator here designed and tested looks perfectible at certain levels, to be retained for the incoming industrial scale prototype.

4. Conclusion

To sum up, a novel MgH_2 tank was proposed to minimize the heat transfer time from the storage material, ensuring fast and efficient sorption conditions. The impact of the heater plate on the response of elemental storage capsule was analyzed through numerical simulations. A stack of capsules and heaters assembled to form a MgH_2-based demonstration tank was tested both numerically and experimentally, leading to original findings as summarized hereafter:

The design of serpentine-shaped heaters and coolers embedded in a steel heat diffuser plays a determining role in reaching rapidly the thermal set-point. Then, in a few minutes, the hydrogen flow can be monitored either by a heating (or cooling) input or via the external pressure set point.

The newly developed SSHS tank design based on stacks of MgH_2 capsules and heat control plates appears to be efficient particularly when it comes to the fast control of hydrogen absorption and desorption processes. In this way, global or partial (sequential) tank implementation can be easily scheduled at demand.

CRediT authorship contribution statement

H. El Alem: Writing – original draft, Software, Formal analysis. **Z. Djordjevic:** Investigation. **N. Skryabina:** Writing – review & editing, Project administration, Conceptualization. **D. Fruchart:** Writing – review & editing, Project administration, Conceptualization. **M. Hajji:** Writing – original draft, Visualization, Formal analysis. **M. Balli:** Formal analysis, Conceptualization, Writing – review & editing. **A. Zaabout:** Writing – review & editing, Software, Formal analysis. **O. Mounkachi:** Resources, Project administration, Funding acquisition, Conceptualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Nomenclature

Symbols	
A	Area (m^2)
e	Thickness (m)
l	Length (m)
r	Radius (m)
D	Diameter (m)
R	Electrical resistance (Ω)
I	Electric current (A)
Q	Thermal flux (W)
H_2	Hydrogen
h	Heat transfer coefficient (W/m^2K)
k	Thermal conductivity (W/mK)
T	Temperature (°C)
Cp	Specific heat (J/kg.K)
V	Volume (m^3)
ε	Porosity of metal hydride
ρ	Density (kg/m^3)
Subscripts	
HCE Cu–Ni	Copper–Nickel heater
HD	Heat diffuser
H_2	Hydrogen
MgH_2 D	Magnesium hydride disk
SS	Steel
HCE Ni–Cr	Nickel–Chromium heater
conv, *cond*	Convection, conduction

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(*continued*)

Symbols	
Nu	Nusselt number (heat transfer coefficient convection/conduction)
Re	Reynolds number (the ratio of inertia to viscous forces)

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